

How do anti-discrimination laws affect firm performance and financial policies? Evidence from the post-World War II period⁺

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Abstract

We exploit the staggered passage of state-level fair-employment laws in the post-World War II period to examine how stronger worker protection against racial discrimination affects firm profitability and financing decisions. We find that firms experience a decline in operating profitability after the passage of anti-discrimination laws. We also document that the adoption of these laws leads to a reduction in debt ratios, which suggests that firms are able to partially offset the negative effects on operating profitability by adjusting their capital structure. Consistent with theoretical predictions, the above effects of anti-discrimination laws are more pronounced for firms in states and cities with a greater proportion of African Americans, firms with high labor intensity, firms in states experiencing high African American migration, and firms in concentrated industries. Some of our evidence suggests that anti-discrimination laws lead to a reduction in statistical discrimination or monopsony power rather than taste-based discrimination. Taken together, our results are supportive of theoretical models of discrimination and show that racial discrimination laws significantly affect firm operating performance and financial policies.

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1. Introduction

In recent years, studies have shown that societal changes can have a large impact on economic outcomes. For example, Huber, Lindenthal, and Waldinger (2021) show that increased discrimination against Jews in the 1930s reduced the value of German firms. Moser, Voena, and Waldinger (2014) find that Jewish immigrants who fled to the United States in the 1930s and 1940s had a positive impact on innovation. In a similar vein, Cohen, Gurun, and Malloy (2017) show that the U.S. policy of forced relocation of people of Japanese descent through internment camps during World War II had lasting effects on international trade. In this paper, we examine how major social change in the U.S. brought about by the Civil Rights Movement impacted firms. In particular, we examine how state-level anti-discrimination laws covering employment practices, put in place following World War II, affect firm-level outcomes such as performance and financial policies.

The focus of the Civil Rights Movement was to obtain equal rights and treatment for all Americans. A major achievement of this movement is the passage of the federal U.S. Civil Rights Act of 1964. Title VII of the Civil Rights Act prohibited discrimination by employers on the basis of race, color, religion, and national origin. However, in the two decades prior to the passage of the Civil Rights Act, civil rights leaders influenced many states to adopt fair-employment laws (Collins, 2003a, 2003b). These laws, often referred to as anti-discrimination laws, were the first generation of U.S. state laws designed to protect workers from racial discrimination in private employment.¹

The state of New York was the first to pass an anti-discrimination (AD) law banning private sector discrimination through the Ives-Quinn Bill in 1945 (e.g., Berger, 1950). This was followed soon after by New Jersey, Massachusetts, Connecticut, Oregon, Rhode Island, New Mexico, and Washington. By the end of 1964, twenty-two non-southern states adopted such laws. While the protections offered by these state-level laws varied across states, the enforcement methods were, for the most part, similar as they were based on New-York's law (e.g., Collins, 2003a; Collins, 2003b). Typically, these laws were backed by state courts and enforced by state agencies who would investigate employment discrimination complaints. Survey based evidence shows that the agencies wielded significant enforcement powers (e.g., Berger, 1950). Moreover, AD laws covered a wide array of employment aspects such as compensation and promotion decisions, worker benefits, working conditions, and hiring and firing practices. The staggered passage of these laws gives us plausibly exogenous variation in the costs and benefits of discrimination for firms.

¹ The idea that the government should not discriminate in employment is embodied in the U.S. constitution through the Fifth and Fourteenth Amendments, which limit the right of government to discriminate in employment practices. However, discrimination in private employment was not restricted by these Amendments.

We first generate hypotheses on the effects of state-level AD laws on firm outcomes. Our hypotheses, explained in detail in Section 2, are primarily motivated by the economic theories of discrimination. Specifically, researchers argue that discrimination can exist in the workplace because (i) employers or employees have a taste to discriminate or animus against minority workers (e.g., Becker, 1957), (ii) risk-averse employers discriminate against minority workers for statistical reasons since signals of their productivity are noisier (e.g., Arrow, 1973; Phelps, 1972), or (iii) employers price discriminate or exercise monopsony power against minority workers (e.g., Robinson, 1933; Becker, 1957). Discrimination can increase firm profits as minority workers receive lower wages compared to equally productive majority workers, thereby reducing the total wage bill of the firm. Discrimination can also benefit firms if done for statistical reasons since it is the optimal resource allocation decision when firms face uncertainty (e.g., Lundahl and Wadensjo, 1984). However, the costs of discrimination include losses of efficiency when employer tastes to discriminate lead to inefficient resource allocation, low morale of minority workers in a segregated environment, and potential legal liability due to discrimination claims (e.g., Northrup, 1944; Collins, 2003a; Collins, 2003b; and Ashenfelter, 1972).

Therefore, *ex ante*, it is unclear how AD laws will affect firm outcomes. As argued above, AD laws eliminate the wage differential between minority and majority workers (Collins, 2003a; Collins, 2003b), which increases salary expenses for firms. In addition, the increased outreach and education programs to integrate majority and minority workers will likely create more costs for firms (Berger, 1950). These higher costs will increase operating expenses of firms thereby reducing operating profitability. If discrimination is statistical in nature, then deviations from optimal resource allocation due to law passage will reduce firms' efficiency and reduce operating profitability. Similarly, reductions in monopsony power after AD law passage will dampen firms' operating profitability. Thus, these factors predict a decline in operating profitability as a result of AD laws. However, if discrimination is taste-based as theorized in Becker (1957), then laws preventing discrimination would enable firms to improve their resource allocation decisions, thereby increasing productivity. According to Becker's arguments under taste based discrimination, productivity improvements likely dominate increases in labor costs, thus predicting a positive effect of AD laws on operating profitability. Overall, there are opposing theoretical predictions on the impact of AD laws on operating profitability, thus motivating an empirical analysis.

We test these competing predictions by examining how changes in state-level anti-discrimination laws affect measures of firm profitability. In all our estimations, we control for a variety of firm-level financial characteristics as well as state-level time-varying economic conditions. Our regressions include firm and year fixed effects to control for time invariant omitted variables at the firm- and year-level, respectively. Additional specifications control directly for the possibility of state-level time trends in the

data. In our baseline regressions, we find that the adoption of AD laws leads to a decrease in operating profitability, with coefficient estimates ranging from -0.7% to -1.1%. Compared to the mean in our sample, this corresponds to a decrease of 3.9% to 6.1%, which suggests that the decline is economically significant. A separate analysis of abnormal stock market returns shows moderately negative effects on firms around the passage of the laws by their states, suggesting that AD law adoption is not only costly for shareholders but also partially unanticipated. The negative effects of AD laws on operating profitability and shareholder wealth are consistent with statistical discrimination or monopsony power arguments, which generally predict negative effects on firms. These results are, however, inconsistent with taste-based discrimination arguments that predict positive effects on firms.

Next, we examine the effect of AD laws on net profitability which is measured as net income divided by assets. We find that the adoption of AD laws is weakly associated with declines in net profitability. The negative coefficients are small in magnitude and either marginally statistically significant or not statistically significant. Because a key difference between operating profits and net income is the effect of firm financial policies, this evidence suggests that firms are able to undo some of the negative effects on operating profits by adjusting their capital structure. Further, we examine changes in employment growth around AD law passage and also find negative effects, however the results are only marginally statistically significant. These slower employment growth rates when AD laws are in place suggest that firms might be reacting to tightening labor market regulations by reducing reliance on labor, though we lack strong statistical support to back this claim.

Our next series of tests examines changes in firm capital structure around the passage of AD laws. We expect firms to adjust their financing choices as a response to changing labor market conditions and declining operating profitability after law passage. Financing decisions also more clearly reflect managers' choices than operating profitability, which is dependent on a host of outside factors (e.g., competition, relationships with customers/suppliers, etc.). On the one hand, AD laws can increase the financial distress risk of firms, which motivates them to reduce financial leverage. The increase in financial distress risk could be attributed to (i) the higher operating costs resulting from increased wages and other related costs pertaining to integration of the workforce, (ii) a reduction in operating flexibility of firms since they are unable to adjust employment effectively due to the fear of being subject to discrimination complaints and litigation at state regulatory agencies (Collins, 2003a, 2003b; Norgren and Hill, 1964), and (iii) any losses in efficiency/profitability after law passage if discrimination was statistical or due to monopsony power.

However, alternative arguments predict the opposite - that AD laws enable firms to increase financial leverage. Specifically, if discrimination is taste based then laws designed to prevent discrimination would enable firms to become more efficient as they could plausibly improve their resource allocations

after law passage. If firms are able to improve labor allocative efficiency, we expect an increase in firm profits/market share, which would plausibly increase their debt capacity. This increase in debt capacity would allow firms to increase financial leverage.² In a similar spirit, if AD laws protect minority workers from unfair dismissal, their expected wage premium to compensate for unemployment risk shrinks, which would allow firms to increase their financial leverage (e.g., Topel, 1984; Abowd and Ashenfelter, 1981; Titman, 1984; Berk, Stanton, and Zechner, 2010; Agrawal and Matsa, 2013).

We test these competing predictions by studying how state-level anti-discrimination laws affect firm debt ratios. In our estimations, we control for firm-level financial characteristics and state-level time-varying economic conditions as well as firm and year fixed effects. In some specifications, we control for state-level time trends. Using multiple measures of debt ratios, we find that the adoption of AD laws leads to a decrease in leverage, with coefficient estimates ranging from -1.3% to -2.1%. Compared to the average in our sample, this corresponds to an economically significant decrease of 4.3% to 10.5% in debt ratios. As such, these findings support the view that firms adopt conservative financial policies to offset some of the negative consequences of the increased operating costs and reduced operating flexibility they face after passage of AD laws. The decline in debt ratios is also consistent with a reduction in statistical discrimination or monopsony power after law passage rather than taste-based discrimination.

The adoption of AD laws must be independent of unobservable factors that drive firm outcomes such as profitability and financial policy to justify their use as a quasi-natural experiment. While it is not possible to completely rule out the influence of unobservable factors, our review of the legal environment in the 1940s shows that corporate lobbying and state-level macroeconomic factors were not the driving force behind these laws. In fact, most business groups opposed the laws. Furthermore, the impetus behind their adoption was the widespread discrimination in employment in war industries during World War II, which hampered the country's ability to support the war effort, and the continued discrimination in employment after the war (Lockard, 1968; Berger, 1950).³ We also run tests to investigate the role of observable factors in the adoption of AD laws, following the approach in Collins (2003b). Specifically, we estimate a Cox proportional hazard model and find that proxies for social factors (e.g., religion, demographics, and union membership) are significant predictors of AD law adoption. In contrast, macroeconomic factors (e.g., state-level unemployment rate and real per capita income growth) are

² Our hypothesis implicitly assumes a positive relation between profitability and leverage, consistent with Xu (2012) who finds a positive relation between leverage and profitability following a shock to expected profits. However, some studies show a negative cross-sectional relation between leverage and profitability (e.g., Rajan and Zingales, 1995).

³ For example, in the state of New York, the driving force behind the Ives-Quinn law was the Governor, who in 1941 appointed a commission to investigate discrimination against minority workers who sought jobs in the war industries.

insignificant predictors. Overall, this evidence provides support for using AD laws as a quasi-natural experiment in our setting.

Our results on operating profits and leverage are robust to a variety of alternative specifications such as using industry-year fixed effects, grouping states by Census division and then interacting these divisions with year fixed effects, setting firm characteristics as of the beginning of the sample and interacting them with year fixed effects, and controlling for AD law passage in neighboring states. Further, the results are robust to dropping firms in states located in the South (none of which passed AD laws). In addition, we find that the effects of AD laws on operating profits and leverage are strongest when states have commissions with powers to enforce the laws. Finally, the results are robust to controlling for the presence of right-to-work laws in the state.

We next conduct heterogeneity analyses to examine how our results differ across economic environments where theory predicts varied effects of these laws. These analyses help to rule out potential alternative explanations of our main results because it is unlikely that an omitted variable is selectively correlated with certain labor or product market characteristics that are predicted by theory to have higher prevalence of discrimination. Our first test is based on the percentage of the population in a state that is African American.⁴ For firms located in states with a high African American population, the potential labor pool has a higher incidence of minority workers, and therefore the effects of AD law passage should be more pronounced. Similarly, AD laws should have little to no effect on firms in states with few minorities in the workforce. Consistent with this intuition, we find that the negative relation between AD laws and both operating profits and financial leverage is significant only in states with high African American population. We find no significant effects for firms in states where the African American population is low.

We repeat this test using data on the percentage of the population in the city of a given firm's headquarters that is nonwhite. This test allows us to use variation in the distribution of African American workers within states to overcome concerns that state-level factors drive our results. We find a negative relation between AD laws and both operating profits and financial leverage for firms located in cities with a high nonwhite population. The relation is statistically insignificant for firms in cities with low nonwhite population. These results show that the treatment intensity of AD laws is varied and, in fact, lower when there is likely to be a lower prevalence of African American workers in the workforce. Further, because city-level measures are at a finer level than state-level measures, these tests help to alleviate concerns that state-level trends drive our results.

⁴ We thank the referee for suggesting these tests to us.

In our final set of tests, we conduct the following additional heterogeneity analyses to bolster our identification. First, we expect that labor-intensive firms would experience greater effects of AD laws because these firms employ more workers in their production activities, and as a result are more likely to hire more minority workers, and hence be subject to discrimination claims. Second, we expect that firms in states which experienced high African American migration from the South would experience greater effects of AD laws since these firms are likely to pick workers from a diverse labor pool that includes these migrant workers, potentially increasing the incidence of discrimination. Finally, we expect that firms operating in highly concentrated industries would experience a greater impact of AD laws because these firms are not subject to the competitive market pressures which tend to reduce discrimination (Becker, 1957). Our empirical evidence is supportive of these predictions.

Our paper contributes to the literature which studies the impact of societal changes on economic outcomes (e.g., Moser, Voena, and Waldinger, 2014; Cohen, Gurun, and Malloy, 2017; Huber, Lindenthal, and Waldinger, 2021). From a historical perspective, the Civil Rights Movement and the reduction in discrimination against African Americans in the United States was a defining change in U.S. history. Several studies examine the effects of anti-discrimination laws on labor market outcomes and find positive effects on wages of nonwhite workers (relative to wages of white workers) but no effect on nonwhite unemployment, occupational distribution, and migration patterns (e.g., Landes, 1968; Heckman, 1976; Collins, 2003a; and Neumark and Stock, 2006). Yet we do not fully understand how these anti-discrimination laws affect economic outcomes, and in particular, firm-level outcomes. Our paper fills this gap in the literature by providing the first firm-level evidence on the effects of race based anti-discrimination laws. We show that AD laws lead to a decline in operating performance of firms. We also show that firms respond to declining profitability by adopting more conservative financing policies. While our paper studies an earlier time period, our findings on race-based discrimination are relevant from a contemporary perspective given the recent focus in many academic disciplines in better understanding economic impacts of discrimination.

A recent stream of literature attempts to understand firm-level and economy wide effects of discrimination. For example, Szymanski (2000) finds that market forces did not eliminate race-based discrimination among English soccer club owners from 1978 to 1993. Huber, Lindenthal, and Waldinger (2021) use the Nazi Party election victory in 1933 in Germany as an exogenous increase in taste-based discrimination. They find that German firms which fired Jewish senior managers in the post-election period experienced losses in value. While Huber, Lindenthal, and Waldinger (2021) look at senior managers, our paper examines laws that affect all workers, and primarily affect lower-ranking workers. Further, some of our evidence is consistent with the view that discrimination in the U.S. was statistical in nature or through

an exercise of monopsony power, although we cannot completely rule out taste-based discrimination. Our work is also related to Hsieh, Hurst, Jones, and Klenow (2019) who build a general equilibrium model and find that a large proportion of U.S. GDP growth from 1960 to 2010 is explained by reduced occupational barriers for minorities and women in accumulating human capital along with declining labor market discrimination. While their focus is on the economy wide effects on GDP, we focus on firm level outcomes and policy choices.⁵

Our paper also contributes to the literature on the determinants of capital structure (e.g., Titman and Wessels, 1988; Rajan and Zingales, 1995; Lemmon, Roberts, and Zender, 2008; Graham and Leary, 2011; Graham, Leary, and Roberts, 2015). In related work, Agrawal and Matsa (2013) find that firms adopt more aggressive financial leverage policies when there is a reduction in unemployment risk of employees. Serfling (2016) shows that firms that experience exogenous increases in worker firing costs, due to the adoption of wrongful discharge laws by states, adopt more conservative financial policies. Similarly, Simintzi, Vig, and Volpin (2014) show a negative relation between debt ratios and a country-level employee protection index. We contribute to this literature by exploiting the staggered adoption of state-level anti-discrimination laws and showing that firms adopt conservative financial policies when they face a rigid labor market that affords stronger rights to employees against racial discrimination.

2. Institutional background, theoretical framework, and hypotheses development

2.1. Institutional background of anti-discrimination laws

A major effort to promote equal job opportunity was the enactment of the Fair Employment Practice Committee (FEPC) in 1941. The committee was set to implement Executive Order 8802 by President Roosevelt which banned discriminatory employment in federal agencies and firms engaged in World War II related work. While the committee was considered a step in the right direction by civil rights advocates, it was largely ineffective since it encountered a lack of political will to move forward and faced administrative hurdles such as a lack of budget and inadequate staffing needs.⁶ Following the war, several states, largely encouraged by civil rights leaders and religious groups, began enacting anti-discrimination laws. Researchers argue that there were two main reasons for states pushing for these laws (e.g., Lockard,

⁵ Dougal, Gao, Mayew, and Parsons (2019) provide evidence of racial discrimination in municipal bond markets. Other papers study discrimination against women focusing primarily on periods in the 1990s and 2000s. Field, Souther, and Yore (2020) find evidence of discrimination against minority and female corporate directors. Weber and Zulehner (2014) find that Austrian firms that discriminate against women have shorter survival rates. Kawaguchi (2007) finds benefits for Japanese firms that discriminate against women. Hellerstein, Neumark, and Troske (2002) find that U.S. plants with market power increased profits by discriminating against women. Gao and Zhang (2017) show that U.S. state laws prohibiting discrimination based on gender identity and sexual orientation, passed since 1977, spur innovation.

⁶ Collins (2001) argues that the FEPC accelerated the pace of economic progress of African Americans.

1968; Berger, 1950). First, World War II produced a heavy demand for labor especially in war related industries. However, despite this heavy demand there was increasing evidence of discrimination against minority workers. Second, there were several instances of severe racial tension in major cities such as Detroit, New York, and Chicago that lead to riots, bombings, and attacks on African American families.

Under this backdrop, New York Governor Thomas Dewey appointed a commission in 1941 to study discriminatory practices in war industries. After its study of discrimination, which lasted multiple years, the commission concluded that racial discrimination was too serious a menace to society to be neglected. It advocated a combination of effective legislation and enforcement along with education and outreach programs to eliminate discriminatory employment practices. The New York State Legislature implemented the suggestions of the commission through the Ives-Quinn Law in 1945, which was the first state-level anti-discrimination law dealing with private employment. Taking their cue from New York, over twenty states appointed similar commissions to investigate employment discrimination and violations of civil rights.⁷ See Table 1 for a list of states that passed anti-discrimination laws.

While the provisions offered by these state-level laws varied across states, the enforcement methods were, for the most part, similar as they were based on New York's law (e.g., Collins, 2003a, 2003b). In most instances, the laws were backed by state courts and enforced by state government appointed agencies who would investigate employment discrimination complaints (e.g., Berger, 1950). The agencies were given considerable enforcement powers including the power to investigate discrimination complaints and to eliminate unlawful discrimination by conciliation, and when necessary, by publicly humiliating the firm or by issuing cease and desist orders on the firm or labor union (e.g., Collins, 2003a, 2003b; Norgren and Hill, 1964).

Survey based evidence shows that the commissions indeed wielded considerable power. For example, Berger (1950) reports that the New York Commission dealt with 787 different respondents during 1945-1949, of whom it found that 65% practiced some form of illegal discrimination. The illegal practices were addressed by the commission via conciliation. Norgren and Hill (1964) argue that fair employment legislation had some effect, particularly in New York. There were 3,262 "violations cleared up" in New York from 1945-1960. Numbers for other states from a similar time period are: 701 for New Jersey, 506 from Connecticut, and 793 from Massachusetts (Landes, 1968, page 547). Collins (2003a) cites several studies that claim substantial improvement for minority groups. As one example, on page 249 it states "The

⁷ Anti-discrimination laws covered a broad range of employment practices for both existing and prospective employees such as compensation and promotion decisions, worker benefits, working conditions, and hiring/termination decisions. In related work, Autor, Donohue, and Schwab (2004, 2006) and Acharya, Baghai, and Subramanian (2014) study wrongful discharge laws passed during 1967-1995, which protected existing employees from unfair dismissal.

New York State Commission against Discrimination reported that 85% of its investigations in 1951 led to “definite improvement” in minority employment patterns.” Some evidence, however, suggests that AD laws were ineffective. For example, Hill (1964) cites evidence that unemployment rates for minority workers were persistently higher than those for white workers in states with AD laws. Newspaper articles from the 1940s and 1950s present arguments both for and against the effectiveness of these laws. It should also be noted that many pro-business groups lobbied against these laws, which offers some evidence that the laws were not passed due to corporate lobbying.

Several studies empirically examine the real effects of race-based anti-discrimination laws on minority workers. Real effects such as wages, annual incomes, unemployment rates, occupational distribution, and migration patterns are examined. In general, the outcomes are measured for nonwhites (or African Americans) relative to whites. Studies show that the passage of anti-discrimination laws had a positive effect on relative wages of nonwhites (e.g., Landes, 1968; Heckman, 1976; Collins, 2003a; and Neumark and Stock, 2006), no effect on nonwhite unemployment (e.g., Landes, 1968; Collins, 2003a; and Neumark and Stock, 2006), and no significant effect on occupational distribution and migration patterns of nonwhites (e.g., Landes, 1968; and Collins, 2003a). Overall, the evidence presented in the above studies would suggest that the wage differential between minority and majority workers declined after passage of the laws, and there was no effect on the employment rates of minority workers after law passage.

One concern with our empirical setting is that states may pass AD laws in response to changing economic conditions in the state.⁸ For example, voters in states where employment and profits are on a downward trend may put more pressure on their legislators to reform the economic system. It is also plausible that voters in states that are doing well economically may be more willing to pass AD laws in order to ensure that economic gains are shared more widely. We estimate Cox proportional hazard models to investigate the role of economic conditions in the timing of AD law passage. We begin with the state-level demographic and economic characteristics studied in Collins (2003b). We add to our specifications variables capturing state-level economic conditions such as overall unemployment rate as well as growth in real per capita income. A detailed description of the hazard model is provided in Internet Appendix A and the estimation results are presented in Internet Appendix Table 1. We find that unemployment rate and the growth in real per capita income are not significantly related to the timing of AD law adoption. The effects of demographic variables on AD law adoption is generally consistent with those found in Collins (2003b). Specifically, we find that states with higher Jewish and Catholic affiliation adopt AD laws faster while states with a higher percentage of Black population are slower to adopt AD laws. Using data from

⁸ We thank the referee for encouraging us to explore this possibility in more detail.

Troy (1957), we find that states with higher affiliation to AFL and CIO unions adopt AD laws faster. Overall, the results from our hazard model estimations suggest that social factors, rather than state-level economic conditions are the main drivers of AD law adoption. It thus appears that the timing of AD law passage is unrelated to firm-level outcomes such as profitability and financial policies, which provides support for using these laws as a quasi-natural experiment to study firm level outcomes.

2.2. Theoretical framework and hypotheses development

Becker (1957) presents the first economic theory for discrimination in the workplace. In his model, employers or employees have a taste to discriminate or animus against minority workers. Taste in the model is the disutility experienced by the majority workers when associated with members of a minority group, which in turn leads them to discriminate against minority members. If managers are discriminating, then minority workers have to compensate discriminating employers for their disutility by offering services at a lower wage compared to equally productive workers from the majority group. This leads to a wage differential between minority and majority workers. Becker (1957) argues that while the wage differential can potentially reduce input costs to firms, discrimination leads to inefficient outcomes to the firm as the managers, who cater to their tastes, end up sub-optimally allocating across the factors of production. Becker (1957) also argues that if employees of the majority group discriminate against minority workers, it would lead to racial segregation in the workforce and have harmful effects on employee morale and productivity.

Alternatively, statistical theories of discrimination argue that discrimination does not arise due to a taste to discriminate but rather it is an outcome of imperfect information about the workers belonging to the minority group. For example, Phelps (1972) and Arrow (1973) argue that risk-averse employers discriminate against minority workers since the variance of their productivity signals is higher than that of the majority workers. The prediction from the models is that workers belonging to the group with less reliable information, usually the minority group, receive lower wages. Finally, researchers argue that employers price discriminate against minority workers because they have monopsony power in the labor market (e.g., Robinson, 1933; Becker, 1957). If this is the case, we would still observe the wage differential but employers would efficiently allocate among factors of production. It is important to note that discrimination could be beneficial to firms if done for statistical reasons (since it is the optimal resource allocation decision when firms face uncertainty) or if done when exercising monopsony power (as firms are able to extract rents from employees) (e.g., Lundahl and Wadensjo, 1984).

The first firm-level outcome we focus on is operating profitability of firms. If anti-discrimination laws curb discrimination in private employment, then we can expect the following negative effects on operating profitability. First, wage expenses would go up since AD laws would eliminate the wage

differential between minority and majority workers (e.g., Landes, 1968; Heckman, 1976; Collins, 2003a; and Neumark and Stock, 2006). Second, anti-discrimination laws would force integration in the workplace, by for example, requiring firms to invest in outreach and education programs that integrate majority and minority workers, which would create new operating costs for firms (Berger, 1950). Third, there is a potential increase in litigation risk and/or public outcry against firms due to enhanced enforcement by state anti-discrimination agencies which would limit the operating flexibility of firms. Therefore, firms worried about facing discrimination lawsuits might not be able to adjust employment effectively after AD law passage. Finally, if discrimination is for statistical reasons or due to monopsony power, firms will experience efficiency losses due to resultant deviations from optimal resource allocation or loss of monopsony power.

We, however, note that there can also be beneficial effects of AD laws on operating profitability. For example, if discrimination is taste based, then AD laws enable firms to efficiently allocate between factors of production, especially when related to the hiring of minority workers. Based on the arguments in Becker (1957), under taste based discrimination, productivity improvements likely dominate the increase in labor costs, which would indicate a positive effect on operating profitability. From a theoretical perspective, the effect of AD laws on operating profitability is, therefore, ambiguous. On the one hand, if discrimination is statistical in nature or due to an exercise of monopsony power, then the resulting efficiency losses after AD law passage would increase operating costs and thereby dampen firms' operating profitability. On the other, if discrimination is taste-based, then laws preventing discrimination would enable firms to improve their resource allocation decisions, which improves operating profitability after law passage.

We also examine changes in financial leverage around AD law passage. We argued earlier that firms would experience losses in efficiency/profitability after AD law adoption if discrimination was statistical or due to monopsony power. We also argued that firms would operate under a higher cost structure and experience a reduction in operating flexibility after AD law passage. The higher cost structure and reduced operating flexibility lead to an increase in financial distress risk of firms, which would induce them to reduce financial leverage. In contrast, if discrimination is taste based, then laws designed to prevent discrimination would improve resource allocation decisions of firms and thereby increase profitability. This improved profitability/efficiency leads to an increase in debt capacity of firms to which they respond by taking on more financial leverage. An alternate literature suggests that the expected wage premium to compensate employees for potential job losses or unemployment risk shrinks if AD laws protect workers from unfair termination (e.g., Topel, 1984; Abowd and Ashenfelter, 1981; Hamermesh and Wolfe, 1990).

We predict that the reduction in labor market risk of workers would enable firms to sustain higher debt levels in their capital structure. Overall, there are opposing effects of AD laws on financial leverage.

3. Data, sample selection, and baseline empirical results

3.1. Anti-discrimination laws

We identify the year that states passed anti-discrimination laws from four different sources: Collins (2003a), Collins (2003b), Neumark and Stock (2006), and Landes (1968). Table 1 displays the year of adoption of anti-discrimination laws by state. We exclude states if Landes (1968) claims that commissions to enforce AD laws did not exist and states in which there is no consensus on the year of adoption of AD law among the sources.⁹ We use the location of firms' headquarters to determine if an AD law is in place or not. For firms located in states that passed a law, the dummy variable *AD Law* is set to 1 for the fiscal year during which the law is passed and for all subsequent years. For prior fiscal years, *AD Law* is set to 0. For the firms located in states that did not pass an anti-discrimination law during our sample period, *AD Law* is set to 0 for all fiscal years. This includes firms located in thirteen states, mostly from the South.

3.2. Sample of firms

Our sample includes publicly traded firms over the time period 1946 to 1964. This time period begins after World War II and ends with the passage of the Civil Rights Act. We begin with a sample of non-financial, non-utility Compustat firms. An issue with Compustat is a selection bias towards larger, historically successful firms prior to 1962 (e.g., Fama and French, 1992; Davis, 1994). To overcome this bias, we expand our sample to include firms on CRSP with financial information available in Moody's Manuals. Specifically, we begin with all firms in CRSP between 1947 and 1967. We remove all firms listed in Compustat and we remove firms with only a single annual observation.¹⁰ This process leaves us with 981 firms. Due to the time intensive nature of hand collecting firm financial information, we sort these firms by name and keep every third observation (327 firms). Of these firms, we are able to hand collect reliable financial information for 169 non-regulated firms covering 1,359 firm-years from Moody's Manuals. Our final sample is 13,633 firm-year observations from 1,652 firms.

Table 2 shows the distribution of firm-years by year, Census region, and state. We see that 42% of observations are from firms located in the Northeast and 31% from the Midwest. Only a relatively small proportion of observations are from the West (11%), reflecting the relative lack of development during this

⁹ These states are Alaska, Delaware, Iowa, Idaho, Maine, Montana, Oklahoma, Vermont, and West Virginia, as well as the District of Columbia. In Section 3.9 we relax these restrictions and find that our inferences remain unchanged.

¹⁰ We require at least two years of data in order to estimate firm fixed effects regressions.

time period. Southern firms comprise 16% of our sample, and we note that no firms located in the South adopted AD laws. New York (17%) and Illinois (11%) represent the most observations in the sample.

3.3. Variables and summary statistics

Our primary measure of firm profitability is *Operating Profitability*, which is operating income scaled by book value of total assets (Compustat item *OIBDP* divided by *AT*). Our measure of net profitability is *Return on Assets* which is the ratio of net income to book value of assets (*NI* divided by *AT*). *Emp. Growth* is the one-year growth rate in total number of employees. In Panel A of Table 3 we find that the mean value of *Operating Profitability* is 0.18 and that of *Return on Assets* is 0.07, both of which indicate that our average sample firm is profitable. The mean value of *Emp. Growth* is 0.06. We examine three main measures of leverage: *Book Leverage*, which is total interest bearing debt divided by book value of total assets (*DLC* plus *DLTT* divided by *AT*), *Total Leverage*, which is the ratio of total liabilities to book value of assets (*LT/AT*), and *Market Leverage* which is calculated as $((DLC+DLTT) / (DLC+DLTT+CSHO*PRCC_F))$. The mean value of *Book Leverage* is 0.17, while mean *Total Leverage* is 0.37. The mean of *Market Leverage* is 0.19 which is available only for 7,704 observations as the stock price data for firms trading on regional stock exchanges is not available on Compustat or CRSP.

We find that the mean value for *Total Assets* is \$167.4 million (in 1960 dollars). The mean value for *Fixed Assets* is 0.35 and firms pay a dividend to common shareholders in 82% of the observations. The average value of *Employees* is 11,820. To capture economic growth at the state level, we measure the percentage increase in real per capita personal income, *State Income Growth*, and the mean is 2.54% (state-level GDP measures are not available during this time period). All variables are defined in the Appendix.

3.4. Abnormal stock returns around passage of AD law

In Panel B of Table 3, we examine the stock market reaction to the passage of AD laws. We read news articles and ascertain relevant dates of legislative action for the passage of the law in eight states with a large number of firms (NY, IL, CA, PA, OH, CT, NJ).¹¹ For each firm in the state, we use a standard event study methodology to cumulate abnormal returns over the (-1,+3) trading day window centered on the date of legislative action. Using trading data for 369 firms across the eight states, we find a mean return of -0.43%, statistically significant at the 5% level (p-value=0.02). Further, we find negative, on average, and statistically significant returns for firms four states (NY, CT, OH, and NJ) and positive, but statistically insignificant returns for firms in the other four states (IL, MA, PA, CA). The negative wealth effects suggest that shareholders, on average, experience losses on announcement of AD laws. It also bolsters the argument

¹¹ We did not include Michigan in this analysis as the exact date of law passage could not be determined.

that the passage of AD laws is partially unanticipated by the stock market, which supports a case to use them as a quasi-natural experiment. Finally, the negative effects of AD laws on shareholder wealth are consistent with either (i) the additional costs borne by firms after reductions in statistical discrimination or (ii) the loss of monopsony power after law passage. The negative sign is generally inconsistent with taste-based discrimination models which would predict that firms would benefit due to superior resource allocation.

3.5. Relation between AD laws and firm profitability

In this section, we use a difference-in-differences methodology to estimate the effect of anti-discrimination laws on firm profitability and employment growth (see e.g., Bertrand and Mullainathan, 2003; Agrawal and Matsa, 2013). Our main regression specification takes advantage of the panel nature of our data, and allows us to include firm fixed effects to difference out any time-invariant firm characteristics. Specifically, we estimate the model:

$$\text{Firm profitability}_{i,s,t} = \beta_1 * \text{AD Law}_{s,t} + \beta * \text{Controls} + \alpha_i + \nu_t + \varepsilon_{i,s,t}, \quad (1)$$

where the dependent variable is a measure of firm profitability for firm i at time t in state s . The coefficient of interest is β_1 , which estimates the effect of AD laws on firm profitability. All specifications include firm fixed effects (α_i) and time fixed effects (ν_t). In our regressions, we control for other determinants of profitability such as firm size (natural logarithm of *Total Assets*), fixed assets (*Fixed Assets*), and dividend payer dummy (*Dividend Payer*). We also control for state-level, time-varying macroeconomic conditions by including *State Income Growth*. Finally, in some specifications we include a state-specific time trend variable to control for any state-level time trends in the data.

The results for these estimations are presented in Table 4. In each panel, Column 1 contains *AD Law* as the only independent variable and column 2 adds *Log (Total Assets)*, *Fixed Assets*, *Dividend Payer*, and *State Income Growth* as control variables. Column 3 includes the above control variables along with a state-specific time trend. We examine the relation between AD laws and firm operating profitability (*Operating Profitability*) in Panel A of Table 4. In all three columns we find that the coefficient on *AD Law* is negative and significant at least at the 5% level. This shows that anti-discrimination law passage is associated with reduced operating profitability of firms. Coefficient estimates reveal that the decreases in operating profitability due to AD law passage are economically significant, ranging from 3.9% - 6.1% for the average firm in our sample.¹² The negative effect of AD laws on operating profitability is consistent

¹² For example, in Column 3 of Panel A, the coefficient on *AD Law* is -0.011, which is 6.1% of the sample mean of 0.18 (0.011/0.18=0.061).

with statistical discrimination or monopsony power and inconsistent with taste-based discrimination models.

In Panel B, we examine the effect of AD laws on net profitability (*Return on Assets*). Net profitability reflects not only the changes to operating profitability as a result of AD laws but also the financial policies chosen by managers in response to these laws. In all three columns we document negative coefficients on *AD Law*, but they are smaller in magnitude and statistically significant at the 10% level only in Column 3. This evidence suggests that firms are able to undo some or most of the negative effects on operating profits by adjusting their capital structure. Finally, in Panel C, we examine how firms adjust labor usage when facing AD laws by studying the relation between firm level employment growth (*Emp. Growth*) and *AD Law*. We document negative coefficients on *AD Law* in all three specifications, however only the coefficient in Column 2 is statistically significant. Therefore, we only find weak evidence of an association between AD laws and employment growth.

3.6. Robustness tests on the relation between AD laws and operating profitability

In Table 5, we conduct several robustness tests on the negative relation between AD laws and operating profitability. We focus only on operating profitability since we find that the effect of AD laws on net profitability and employment growth is muted. First, to investigate if industry-specific trends are driving the results we run specifications where we include two digit SIC-year fixed effects (Column 1). Second, to investigate if our results are driven by regional effects, we include Census division-year fixed effects (Column 2). In order to run this test, we link each state in our sample to one of the nine Census divisions. Third, to alleviate concerns that our control variables are also affected by AD laws and are therefore endogenous, we remove time-varying measures of firm characteristics. Instead, all firm characteristics are measured in a baseline year – the first year of data – and then interacted with year fixed effects (Column 3). We consider all beginning characteristics from Table 4 including natural logarithm of *Total Assets*, *Fixed Assets*, *Dividend Payer*, and *State Income Growth*. Finally, the effect of the anti-discrimination laws should end at the borders of the state, so similar laws in neighboring states should not be a significant determinant of firm level outcomes for firms headquartered in a state. Consistent with the approach in Agrawal and Matsa (2013), we compute the median value of *AD Law* based on the states which share a border with the state where the firm's corporate headquarters are located and include this value (*AD Law Neighboring*) as a control (Column 4).

In Table 5, we examine the relation between AD laws and operating profitability of firms (*Operating Profitability*) based on the above specifications. In all four columns, we find that the coefficient on *AD Law* is negative and statistically significant at least at the 5% level. This shows that the negative

relation between anti-discrimination laws and operating profitability of firms survives a variety of robustness tests. In Column 4, we find that the coefficient on *AD Law* is statistically significant but the coefficient on *AD Law Neighboring* is statistically insignificant. Therefore, our results continue to hold even after controlling for neighboring states' anti-discrimination law status, which is inconsistent with the view that regional macroeconomic conditions are driving our results.

3.7. Relation between AD laws and financial leverage

In this section, we investigate the effect of anti-discrimination laws on firm capital structure. One benefit of examining capital structure is that it is a policy choice of firm managers and therefore more in their control than operating profitability. Our main regression specification is similar to the one we employed in Section 3.5. Specifically, we estimate the following model:

$$Leverage_{i,s,t} = \beta_1 * AD\ Law_{s,t} + \beta * Controls + \alpha_i + v_t + \varepsilon_{i,s,t}, \quad (2)$$

where the dependent variable is a measure of financial leverage for firm i at time t in state s . The coefficient of interest is β_1 , which estimates the effect of AD laws on firm leverage. Our specification includes firm fixed effects (α_i) and time fixed effects (v_t) to control for time-invariant omitted variables at the firm and year level. In our regressions, we control for factors that are correlated with leverage such as firm size (natural logarithm of *Total Assets*), *Return on Assets*, *Fixed Assets*, and *Dividend Payer*. We control for time-varying state-level macroeconomic conditions by including *State Income Growth*. We also explore specifications which include a state-specific time trend variable to account for state-level trends.

Table 6 presents results from estimating Equation (2). The dependent variable is *Book Leverage* in Panel A, *Total Leverage* in Panel B, and *Market Leverage* in Panel C. All three panels include firm and year fixed effects. In each panel, Column 1 contains *AD Law* as the only independent variable and Column 2 adds *Log (Total Assets)*, *Return on Assets*, *Fixed Assets*, *Dividend Payer*, and *State Income Growth* as control variables. Column 3 includes the above control variables along with a state-specific time trend variable. We find that the coefficient on *AD Law* is negative and statistically significant at least at the 5% level in all specifications in the three panels. For example, in Column 3 of Panel A, where the dependent variable is *Book Leverage*, the coefficient on *AD Law* is -0.014 (t-stat = -2.86) and is statistically significant at the 1% level. This result shows that anti-discrimination laws are negatively associated with firm leverage. We estimate the economic significance and infer that in states where *AD Law* equals 1, *Book Leverage* drops by 8.2% of the sample mean (0.014/0.17). Furthermore, coefficient estimates in Panels A, B, and C show that changes in debt ratios range from 4.3% - 10.5% depending on the specific measure of leverage examined. Overall, this shows that the effect of *AD Law* on leverage is economically significant.

3.8. Robustness tests on the relation between AD laws and financial leverage

In this section, we conduct several robustness tests on the negative relation between AD laws and financial leverage. In Table 7, we examine the relation between AD laws and *Book Leverage* in four alternative specifications that mirror those in Table 5 (see Section 3.6). In Column 1, we include two digit SIC-year fixed effects. In Column 2, we include Census division-year fixed effects. In Column 3, we include beginning characteristic-year fixed effects. Finally, in Column 4, we include the median value of neighboring state AD laws (*AD Law Neighboring*) as a control. We find that the coefficient on *AD Law* is negative and statistically significant at least at the 10% level in all four specifications. In unreported tests, we examine the relation between *AD Law* and the other two measures of leverage (*Total Leverage* and *Market Leverage*) and find that our inferences generally remain unchanged. Overall, we find that the negative relation between anti-discrimination laws and financial leverage of firms generally survives a wide variety of robustness tests.

3.9. Additional robustness tests: Dropping regions/states and enforcement strength of AD laws

In Tables 8 and 9, we conduct several robustness tests. In each panel, the dependent variable is *Operating Profitability* in Column 1 and *Book Leverage* in Column 2. All estimations include appropriate control variables along with firm and year fixed effects and state-specific time trends. The first set of tests controls for concurrent legislation that may affect firm outcomes. In many states the adoption of anti-discrimination laws might have coincided with other labor regulations, one of which is the adoption of right-to-work (*RTW*) laws. In *RTW* states, workers in unionized firms cannot be forced to pay union dues or join unions. Therefore, the power of labor unions is likely to be weaker in *RTW* states. Consistent with this argument, Matsa (2010) shows that firms strategically take on more leverage in non-*RTW* states, where unions are stronger, in order to improve their bargaining position with labor unions. In Panel A of Table 8, we control for the *RTW* vs. non-*RTW* status of firms in our estimations. The coefficients on *AD Law* remain significant the 1% level in both columns. Consistent with Matsa (2010), we find that the *RTW* dummy is negative and statistically significant in the leverage regression in Column 2. Thus, our inferences remain unchanged after controlling for the *RTW* status of firms.

The next set of tests are based on excluding firms from certain geographic locations. Since Southern states did not adopt anti-discrimination laws during our sample period, it is plausible that economic conditions and labor markets in these states are unique compared to the rest of the U.S. Though our estimations survive the inclusion of state-time trends, we nevertheless, conduct additional robustness tests. In Panel B of Table 8, we drop all states located in the South from the analysis and continue to find a negative and statistically significant effect of *AD Law* on *Operating Profitability* and *Book Leverage*. In

Panels C and D, we explore whether firms located in New York or Illinois have an outsized influence on our results, as these states account for 17% and 11% of our sample, respectively. Our inferences generally remain unchanged if we drop firms in these states.

Our main results are based on a sample of firms from states where there is consensus on the year of AD law passage in Collins (2003a, 2003b), Neumark and Stock (2006), and Landes (1968) and states where Landes (1968) argues that the laws provided for commissions with enforcement powers (see Section 3.1). In our next set of robustness tests, we rebuild our sample by relaxing both of these criteria, which increases the sample size. In particular, we use the year of law passage in Landes (1968) for states which lacked consensus, and we also include firms in states which Landes (1968) argues that the commissions had no enforcement powers.¹³ We create a dummy variable *AD Law Augmented* which is set to one if a state anti-discrimination law is in place in the augmented sample, and zero otherwise. All estimations include control variables along with firm and year fixed effects and state-specific time trends. We report the results for this augmented sample in Panel A of Table 9. Our inferences remain unchanged in the augmented sample as we find a negative and statistically significant coefficient on *AD Law Augmented* in both *Operating Profitability* and *Book Leverage* regressions.

Next, we examine variation in the enforcement powers of AD laws. To run this test, we separate *AD Law Augmented* into two dummy variables: *AD Law Enforce* and *AD Law Non Enforce*. *AD Law Enforce* is set to one if an enforceable anti-discrimination law is in place in the augmented sample, and zero otherwise. *AD Law Non Enforce* is set to one for states where anti-discrimination laws are in place, but Landes (1968) argues that the laws lacked enforcement powers, and zero otherwise. In Panel B of Table 9, we run our estimations by including both of these dummy variables. We find that the coefficient on *AD Law Non Enforce* is insignificant which suggests that in non-enforcing states the AD law had no effect on firm level outcomes. Furthermore, we find that the coefficient on *AD Law Enforce* is negative and significant, which suggests that the effect of AD laws on *Operating Profitability* and *Book Leverage* is confined to states where significant enforcement powers existed.

Finally, in Internet Appendix Table 2, we examine the effect of *AD Law* on leverage and operating profitability when adding *Earnings Volatility* as an additional control variable. We also investigate three alternative measures of leverage in this table, *Log Total Debt*, *Net Book Leverage*, and *Net Market Leverage*, which are defined in the table legend. Again, we find a negative and statistically significant (at least at the 10% level) coefficient on *AD Law* in all seven regressions.

¹³ States that meet both of these criteria are Alaska, Delaware, Iowa, Idaho, Maine, Montana, Oklahoma, Vermont, and West Virginia, as well as the District of Columbia.

4. Heterogeneity analyses

In this section, we conduct heterogeneity analyses by sub-dividing our sample of firms based on certain labor market or product market characteristics. These analyses are helpful since it is unlikely that an omitted variable is selectively correlated with the labor or product market characteristic that is predicted by theory to have higher prevalence of discrimination. We then examine if there are differences in the relation between AD laws and firm-level outcomes based on the proxies for labor and product market characteristics.

4.1. Heterogeneity analyses based on minority percent of the population

Our first test is based on the percentage of the population in the state of a given firm's headquarters that is Black. When firms are located in states with a high Black population, the effects of AD laws should be more pronounced since the potential labor pool of these firms has a higher number of Black workers, and as a result there would likely be a larger number of Black workers in their workforce. In contrast, when firms are located in states with a low Black population, there is likely a lower prevalence of Black workers in the workforce, and therefore the effects of AD laws should be muted for these firms. To conduct this test, we collect data on the total population as well as the Black population for each state from the 1950 Census. We then calculate the percentage of a state's population that is Black and split our firm-year panel based on the sample median of this variable.

We report results from the sub-sample test in Table 10. In Columns 1 and 2, the dependent variable is *Operating Profitability* and in Columns 3 and 4 it is *Book Leverage*. All estimations contain relevant control variables as well as firm and year fixed effects along with a state-specific time trend. We find that the coefficient on *AD Law* is negative and significant at the 1% level for firms located in states with above median percent Black population (Columns 1 and 3). For firms in states with below median percent Black population, the coefficient on *AD Law* is negative but not statistically significant (Columns 2 and 4). These results show that the effect of AD laws on operating profits and financial leverage is significant in sub-samples with more minorities in the workforce, which is consistent our prediction.

To exploit heterogeneity at the city-level, we repeat this test using the percentage of the population in the city of a given firm's headquarters that is nonwhite.¹⁴ Since labor markets are likely to be localized, this test allows us to compare the effects of AD laws for firms located in an area with high nonwhite population to those located in the same state but in an area with lower nonwhite population. To run this test, we hand collect data on the city of corporate headquarters for our sample firms as well as the closest

¹⁴ 1950 Census data at the state-level provides Black population statistics, while data at the SMA-level provides nonwhite population statistics.

Standard Metropolitan Area (SMA). We manually link each SMA in our sample to data on the nonwhite population in the 1950 Census of Population Advanced Report “Population, by Color, for Standard Metropolitan Areas and Cities of 50,000 or more”. We obtain the location of the city of headquarters for 1,631 out of 1,653 firms in our sample and we link these cities to an SMA for 1,536 firms. We measure the percentage of population at the SMA level that is nonwhite. We then divide our firm-year panel into high and low groups based on the sample median value of this variable. We illustrate the intuition of this test with two cities in the state of Pennsylvania. We find that the nonwhite population of Philadelphia is more than double that of Pittsburgh (13% vs. 6%). Thus, Pennsylvania’s AD law should have a large impact on firms in Philadelphia and a small impact on firms in Pittsburgh.

In Table 11, we examine the relation between AD laws and firm outcomes for firms located in cities in the high and low sub-samples. The layout of Table 11 is similar to that of Table 10. In the high sub-sample (Columns 1 and 3), we find that the coefficient on *AD Law* is negative and significant at the 1% level in both the *Operating Profitability* and *Book Leverage* regressions. The coefficient on *AD Law* is negative but statistically insignificant in the low sub-sample (Columns 2 and 4). Overall, similar to the findings in Table 10, our results show that the effect of AD laws on operating profits and financial leverage is significant in sub-samples with relatively more minority workers at the city-level. The city-level tests are advantageous since city-level measures are at a finer level than state-level measures, and thus these results are unlikely to be driven by state-level factors.

4.2. Additional heterogeneity analyses

In this section, we conduct heterogeneity analyses based on additional labor and product market characteristics. First, we investigate how the relation between anti-discrimination laws and firm outcomes varies across sub-samples of firms with differing labor intensity. We posit that labor intensive firms are more likely to be subject to discrimination claims since they employ more workers and, hence, more minority workers. Therefore, labor intensive firms will experience a greater increase in operating costs after AD law passage and hence a greater reduction in operating profitability. Furthermore, managers of labor intensive firms are more likely to take into account the increase in operating costs when setting their capital structure, which would suggest that labor intensive firms would also experience a greater reduction in debt ratios after AD law passage.

We measure labor intensity by the ratio of number of employees to sales. Our sample of firms is split based on which firms lie above vs. below the median value of labor intensity. We report the results for these estimations in Panel A of Table 12. We expect the relation between *AD Law* and *Operating Profitability* (or *Book Leverage*) to be more pronounced in the sub-sample of high labor intensity, as proxied

by high employees to sales ratio. The dependent variable is *Operating Profitability* in Columns 1 and 2 and *Book Leverage* in Columns 3 and 4. Columns 1 and 3 (2 and 4) contains estimations for the sub-sample of firms with high (low) labor intensity. Consistent with our conjecture, we find that the coefficient on *AD Law* is negative and statistically significant in the sub-sample of firms with high labor intensity in Columns 1 and 3. In contrast, it is insignificant in the sub-sample of firms with low labor intensity in Columns 2 and 4.

Our second test is based on the wave of African American migration which relocated more than six million African Americans from the Southern states to states in the North, Midwest, and West. Historians have argued that the migration was driven by a lack of economic opportunities and tough segregationist laws in the Southern states, coupled with the rising demand for industrial workers (e.g., Tolnay and Beck, 1990; Tolnay and Beck, 1992, Collins, 1997). In states that experienced a greater inflow of migrant African Americans, firms would have to hire workers from a labor pool with a higher proportion of minority workers. These firms are more likely to be subject to discrimination claims since they employ more minority workers. These firms also likely experience a greater increase in operating costs and hence a greater reduction in operating profitability after *AD law* passage. Managers of these firms are more likely to take into account such costs when setting their capital structure. We, therefore, expect a more negative effect of *AD Law* on operating profitability and debt ratios for firms in states with higher African American influx.

We collect data on African American migration from Professor James Gregory's website.¹⁵ To ensure an overlap with our sample period, we focus on the second wave of migration which led to the movement of 5 million African Americans in the period 1950-1970. Since the focus of this test is on migration from Southern states, we only examine a sample of non-Southern states. Our measure of migration is based on the growth rate in the number of Southern-born African Americans living in each state between 1950 and 1970. We subdivide our sample of firms in non-Southern states based on the sample median for this measure. The results for these estimations are reported in Panel B of Table 12. The layout of Panel B is similar to that of Panel A. Consistent with our conjecture, we find that the coefficient on *AD Law* is negative and statistically significant in the sub-sample of firms with high African American influx (Columns 1 and 3). In contrast, the coefficient on *AD Law* is statistically insignificant in the sub-sample of firms with low African American influx (Columns 2 and 4).

As a final test, we examine the effect of product market competition. Becker (1957) argues that as barriers of entry in an industry decrease, new firms with lower tastes to discriminate can enter the market.

¹⁵ http://depts.washington.edu/moving1/black_migration.shtml

Due to their lower taste to discriminate, these firms can more profitably allocate labor between minority and majority workers and extract market share from the discriminating firms. It is important to note that competitive pressures can dampen statistical discrimination as well, to the extent that market competition puts pressure on employers to develop better tests to measure productivity. As such, the threat of competition would mitigate the tendency to discriminate against minority workers in more competitive product markets. In concentrated product markets, discrimination is more likely to prevail due to the lack of such competitive threats. Some empirical support for this argument is found in Levine, Rubinstein, and Levkov (2014) who show that banking deregulation across U.S. states in the 1970s to 1990s (which increased competition) improved the relative wages of African American workers. Similarly, Donohue and Heckman (1991) argue that firms in many industries, who sought to control labor costs in the face of rising foreign competition, took the pretext of AD law passage to employ more minority workers.

We examine the effect of product market competition by splitting our sample of firms into high vs. low concentration industries. We expect that discriminatory practices would be more prevalent in the high concentration (i.e., low competition) sub-sample prior to the AD law passage. The law passage would then minimize the prevalence of these practices in high concentration industries to a greater extent than in low concentration industries. Thus, we expect to observe a more negative coefficient on *AD Law* in the high concentration sub-sample. We present the results for these estimations in Panel C of Table 12. As a measure for industry concentration, we employ the industry Herfindahl-Hirschman Index (*HHI*) calculated using revenue based market share in the two-digit SIC industry of firms. The layout of Panel C is similar to Panels A and B. Consistent with our conjecture, we find that the coefficient on *AD Law* is indeed negative and significant in the sub-sample of firms in the high industry concentration group (Columns 1 and 3). The coefficient is insignificant in the low concentration sample in Column 2, while it is negative and significant at the 10% level in Column 4.

4.3. Additional tests on covariate balance and timing of firm outcome changes around AD law passage

In this section, we conduct robustness tests based on Serfling (2016). First, we investigate if our results hold once there is covariate balance between our treatment and control firms based on propensity score matching. Treatment firms are those where an *AD Law* was adopted in year t whereas control firms are those that did not. Further details on the matching procedure are provided in Internet Appendix Table 3. The sample for this test includes all observations of the treatment and matched control firms in the five years before and five years after the adoption of the laws. In Panel A of Internet Appendix Table 3 we find that treatment and control firms are similar in terms of firm financial characteristics in year $t-1$. In Panel B, our analysis shows that the effects of the laws on book leverage and operating profits are generally felt by treatment firms in the period following law passage.

We next examine changes in operating profitability and book leverage relative to the timing of the passage of anti-discrimination laws. In Internet Appendix Table 4 we find that the decline in book leverage and operating profitability is driven primarily in the year of adoption of AD law or in the years after the adoption of the law. Furthermore, we find no evidence of declining trends two years prior to the passage of the law. While there is no evidence of a decline in operating profitability in the year prior to law passage, there is some evidence of a reduction in leverage in the year prior to passage of the law. Overall, our results show that changes in firm-level outcomes are predominantly confined to the period around or after the passage of AD laws.

Finally, we perform a falsification test by randomly assigning each state a year of AD law adoption while maintaining the distribution of adoption dates in Table 1. If anti-discrimination laws truly impact firm profitability and leverage, then we should observe no effect on our randomized AD law dummy. Next, we replicate the firm fixed effects panel regressions with *Operating Profitability* and *Book Leverage* as our dependent variables and the randomized AD law dummy as our key independent variable. We repeat this process a total of 100 times and examine the distribution of the coefficient on the AD law dummy, the standard error, and the t-statistic. Internet Appendix Table 5 shows that the mean coefficients, standard errors, and t-statistics are close to zero for *Operating Profitability* and *Book Leverage* regressions. Given this evidence, we conclude that there is no relation between profitability or financial leverage and the randomized AD law variable. This falsification exercise provides further support for the view that the actual adoption of AD laws affects firms.

5. Conclusions

This paper examines the effects of anti-discrimination laws, which address race-based discrimination in private employment, on firm-level outcomes. In the period between the end of World War II in 1945 and the passage of the federal Civil Rights Act in 1964, over twenty states passed the first generation of anti-discrimination laws in the United States. We exploit the staggered passage of these laws to examine how laws that give workers stronger protection against racial discrimination affect firm profitability and financing decisions. While prior research shows that these laws have positive effects on the relative wages of minority workers without negatively affecting employment and occupational distribution (e.g., Landes, 1968; Heckman, 1976; Collins, 2003a; and Neumark and Stock, 2006), there is little empirical evidence on aggregate effects of these anti-discrimination laws, either at the firm or economy level. We fill this gap in the literature by showing that (i) anti-discrimination laws lead to a decline in operating profitability and (ii) firms partially offset the negative effects on profitability by adopting more conservative financing policies.

We conduct heterogeneity analyses to document that the effects of anti-discrimination laws on firm operating profitability and financial leverage are more pronounced in environments where economic theory predicts stronger effects of discrimination. Specifically, we find that the effects are generally stronger for firms in states and cities with a higher African American population, for more labor-intensive firms, for firms in states with greater African American migration from the South, and for firms in more concentrated industries. Some of our results are consistent with the notion that discrimination was statistical or due to the exercise of monopsony power, although we cannot fully rule out taste-based discrimination in our tests. Given our quasi-natural experimental design, our results are suggestive that the effects of AD laws are causal.

Our study is the first to investigate the effects of anti-discrimination laws on firm outcomes such as profitability and capital structure, and one of only a handful of studies to examine firm financial policies during the 1940s to 1960s. From a historical perspective, we know little about how discrimination against minorities affected firm-level outcomes. Recently, many academic disciplines have tried to gain a better understanding of the economic impacts of discrimination, spurred, in part, by the documented persistence of racial disparities.¹⁶ This study can help both researchers and legislators better understand the impacts of anti-discrimination laws on firms and the economy. Thus, our findings are relevant from both an historical and a contemporary perspective.

¹⁶ For recent evidence on racial disparities, see Derenoncourt and Montialoux (2021), Chetty, Henderson, Jones, and Porter (2019), and Bayer and Charles (2018). In recent years, professional associations in various academic disciplines such as economics, finance, marketing, management, and accounting have all held panel discussions at annual meetings on discrimination and/or issued calls for research on discrimination.

Appendix: Variable Definitions

Variable	Construction
<i>AD Law</i>	Dummy variable equal to 1 if a state-level anti-discrimination law is in place, and 0 otherwise.
<i>Operating Profitability</i>	OIBDP / AT
<i>Return on Assets</i>	NI/AT
<i>Emp. Growth</i>	One year growth rate in number of firm level employees
<i>Employees</i>	Number of employees for a firm (EMP)
<i>Book Leverage</i>	(DLC + DLTT) / AT
<i>Total Leverage</i>	LT/AT
<i>Market Leverage</i>	(DLC+DLTT) / (DLC+DLTT + CSHO*PRCC_F)
<i>Total Assets</i>	AT, adjusted with CPI index to 1960 dollars
<i>Fixed Assets</i>	PPENT/AT
<i>Dividend Payer</i>	Dummy variable equal to 1 if common dividends is > 0
<i>State Income Growth</i>	Percentage change in real per capita income in the state

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Table 1: Year of adoption of anti-discrimination laws

Data are from Collins (2003a, 2003b), Neumark and Stock (2006), and Landes (1968). We exclude states where there is no consensus on the year of adoption of anti-discrimination laws or where commissions with enforcement powers were non-existent. States that meet these criteria are Alaska, Delaware, Iowa, Idaho, Maine, Montana, Oklahoma, Vermont, and West Virginia, as well as the District of Columbia.

State	Code	Year	State	Code	Year
Alabama	AL	None	North Dakota	ND	None
Arkansas	AR	None	Nebraska	NE	1965
Arizona	AZ	1965	New Hampshire	NH	1965
California	CA	1959	New Jersey	NJ	1945
Colorado	CO	1957	New Mexico	NM	1949
Connecticut	CT	1947	Nevada	NV	1965
Florida	FL	None	New York	NY	1945
Georgia	GA	None	Ohio	OH	1959
Hawaii	HI	1963	Oregon	OR	1949
Illinois	IL	1961	Pennsylvania	PA	1955
Indiana	IN	1963	Rhode Island	RI	1949
Kansas	KS	1961	South Carolina	SC	None
Kentucky	KY	1966	South Dakota	SD	None
Louisiana	LA	None	Tennessee	TN	None
Massachusetts	MA	1946	Texas	TX	None
Maryland	MD	1965	Utah	UT	1965
Michigan	MI	1955	Virginia	VA	None
Minnesota	MN	1955	Washington	WA	1949
Missouri	MO	1961	Wisconsin	WI	1957
Mississippi	MS	None	Wyoming	WY	1965
North Carolina	NC	None			

Table 2

Distribution of observations by fiscal year, region, and state.

Year	Frequency	Percent
1946	5	0.04
1947	71	0.52
1948	73	0.54
1949	76	0.56
1950	599	4.39
1951	606	4.45
1952	610	4.47
1953	614	4.50
1954	626	4.59
1955	640	4.69
1956	652	4.78
1957	665	4.88
1958	682	5.00
1959	699	5.13
1960	1,139	8.35
1961	1,245	9.13
1962	1,457	10.69
1963	1,596	11.71
1964	1,578	11.57
Total	13,633	100

Region	Frequency	Percent
Northeast	5,693	41.76
Midwest	4,292	31.48
West	1,496	10.97
South	2,152	15.79
Total	13,633	100

State	Frequency	Percent
NY	2,374	17.41
IL	1,508	11.06
CA	1,073	7.87
PA	1,063	7.80
OH	1,028	7.54
CT	924	6.78
NJ	732	5.37
TX	690	5.06
MI	576	4.23
MA	446	3.27
FL	354	2.60
MO	314	2.30
WI	314	2.30
MN	283	2.08
GA	232	1.70
IN	210	1.54
VA	189	1.39
CO	167	1.22
NC	147	1.08
MD	107	0.78
TN	103	0.76
LA	82	0.60
SC	79	0.58
WA	79	0.58
NH	78	0.57
RI	76	0.56
AZ	73	0.54
KY	73	0.54
AL	52	0.38
KS	52	0.38
UT	49	0.36
AR	36	0.26
OR	24	0.18
HI	15	0.11
NV	14	0.10
MS	8	0.06
NE	7	0.05
NM	2	0.01
Total	13,633	100.00

Table 3: Summary statistics

This table presents the univariate statistics of firms in our sample (Panel A) and abnormal stock returns around the passage of AD laws (Panel B). Mean, standard deviation, and percentiles are presented. The sample comprises 13,633 publicly traded firm-years from 1946-1964. Financial and utility firms are not included. *AD Law* is a dummy set to one if a state-level anti-discrimination law is in place, and zero otherwise. *Operating Profitability* is the ratio of Compustat *OIBDP* over *AT*. *Return on Assets* is the ratio of *NI* to *AT*. *Emp. Growth* is the one-year growth rate of the number of firm-level employees. *Book Leverage* is the sum of debt in current liabilities and long-term debt (Compustat *DLC* plus *DLTT*) divided by the book value of assets (Compustat *AT*). *Total Leverage* is total liabilities (Compustat *LT*) divided by the book value of assets (Compustat *AT*). *Market Leverage* is the sum of debt in current liabilities and long-term debt (Compustat *DLC* plus *DLTT*) divided by the market value of assets (Compustat items *DLC+DLTT* + *CSHO*PRCC_F*). All other variables are described in the Appendix. All continuous variables are winsorized at the 1st and 99th percentiles. Abnormal returns are cumulated over the (-1,+3) trading day window centered on dates of legislative action in each state. Abnormal returns are calculated using a market model, with parameters estimated over the 253 day window beginning 30 days prior to the event date (minimum 30 observations required).

Panel A: Summary Statistics

Variable	N	Mean	Std. dev.	p25	Median	p75
<i>AD Law</i>	13,633	0.61	0.49	0.00	1.00	1.00
<i>Operating Profitability</i>	13,275	0.18	0.09	0.12	0.17	0.23
<i>Return on Assets</i>	13,472	0.07	0.05	0.04	0.07	0.09
<i>Emp. Growth</i>	9,681	0.06	0.19	0.00	0.01	0.10
<i>Book Leverage</i>	12,799	0.17	0.15	0.04	0.15	0.26
<i>Total Leverage</i>	12,506	0.37	0.16	0.25	0.36	0.47
<i>Market Leverage</i>	7,704	0.19	0.18	0.03	0.16	0.31
<i>Total Assets</i>	13,633	167.40	402.01	15.00	42.78	131.46
<i>Fixed Assets</i>	13,586	0.35	0.19	0.22	0.32	0.47
<i>Dividend Payer</i>	13,629	0.82	0.38	1.00	1.00	1.00
<i>Employees</i>	11,180	11.82	23.69	1.50	4.00	10.70
<i>State Income Growth (%)</i>	13,633	2.54	2.91	1.14	2.72	4.40

Panel B: Abnormal Returns

State	N	Mean	Std. dev.	Median	p-value
All Eight States	369	-0.43%	3.46%	-0.58%	0.02**
NY	94	-1.51%	2.80%	-1.28%	<0.01***
CT	32	-1.28%	4.18%	-0.65%	0.09*
OH	36	-1.21%	3.35%	-1.37%	0.04**
NJ	18	-1.00%	2.06%	-0.84%	0.05*
IL	78	0.15%	3.51%	-0.41%	0.70
MA	30	0.36%	2.24%	0.32%	0.39
PA	45	0.51%	3.46%	-0.30%	0.33
CA	36	1.16%	4.54%	0.20%	0.13

Table 4: Effects of AD laws on profitability and employment

Ordinary least squares regressions for a sample of public firms from 1946 to 1964. *AD Law* is a dummy set to one if a state-level anti-discrimination law is in place, and zero otherwise. The dependent variable in Panel A is *Operating Profitability* which is the ratio of Compustat *OIBDP* over *AT*. In Panel B, the dependent variable is *Return on Assets* which is the ratio of *NI* to *AT*. In Panel C, the dependent variable is *Emp. Growth* which is the one-year growth rate of the number of firm-level employees. All other variables are described in the Appendix. All regressions include firm fixed effects and year fixed effects. Column 3 also includes state-specific time trends. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

	(1)	(2)	(3)
<i>Panel A: Operating Profitability</i>			
Dep. variable	<i>Operating Profitability</i>	<i>Operating Profitability</i>	<i>Operating Profitability</i>
<i>AD Law</i>	-0.008** (-2.175)	-0.007** (-2.191)	-0.011*** (-3.214)
<i>Log (Total Assets)</i>		0.006 (1.421)	0.005 (1.267)
<i>Fixed Assets</i>		-0.140*** (-6.670)	-0.138*** (-6.731)
<i>Dividend Payer</i>		0.026*** (5.735)	0.026*** (5.670)
<i>State Income Growth</i>		0.002*** (4.914)	0.002*** (5.197)
Firm and year fixed effects	Yes	Yes	Yes
State-specific time trends	No	No	Yes
Observations	13,266	13,235	13,235
R-squared	0.69	0.70	0.71
<i>Panel B: Return on Assets</i>			
Dependent variable	<i>Return on Assets</i>	<i>Return on Assets</i>	<i>Return on Assets</i>
<i>AD Law</i>	-0.001 (-0.750)	-0.001 (-0.872)	-0.003* (-1.780)
<i>Log (Total Assets)</i>		0.003 (1.454)	0.003 (1.461)
<i>Fixed Assets</i>		-0.096*** (-8.891)	-0.096*** (-8.826)
<i>Dividend Payer</i>		0.019*** (7.654)	0.019*** (7.631)
<i>State Income Growth</i>		0.001*** (4.030)	0.001*** (4.261)
Firm and year fixed effects	Yes	Yes	Yes
State-specific time trends	No	No	Yes
Observations	13,472	13,453	13,453
R-squared	0.64	0.66	0.67

continued...

Table 4, continued

	(1)	(2)	(3)
<i>Panel C: Employment Growth</i>			
Dep. variable	<i>Emp. Growth</i>	<i>Emp. Growth</i>	<i>Emp. Growth</i>
<i>AD Law</i>	-0.011 (-1.584)	-0.012* (-1.756)	-0.008 (-0.987)
<i>Log (Total Assets)</i>		0.029** (2.333)	0.038*** (3.077)
<i>Fixed Assets</i>		-0.106** (-2.247)	-0.111** (-2.347)
<i>Dividend Payer</i>		0.003 (0.213)	0.008 (0.603)
<i>State Income Growth</i>		0.004*** (3.830)	0.004*** (3.999)
Firm and year fixed effects	Yes	Yes	Yes
State-specific time trends	No	No	Yes
Observations	9,545	9,537	9,537
R-squared	0.26	0.27	0.28

Table 5: Robustness tests on the effects of AD laws on operating profitability

Ordinary least squares regressions for a sample of public firms from 1946 to 1964. *AD Law* is a dummy set to one if a state-level anti-discrimination law is in place, and zero otherwise. The dependent variable is *Operating Profitability* which is the ratio of Compustat *OIBDP* over *AT*. Column 1 includes two digit SIC-year fixed effects, column 2 includes Census division-year fixed effects, and column 3 includes beginning characteristic-year fixed effects. Column 4 includes *AD Law Neighboring*, which is the median value of *AD Law* status based on neighboring states. All regressions include firm fixed effects and year fixed effects as well as controls as in Table 4. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

Dependent variable	(1)	(2)	(3)	(4)
	<i>Operating Profitability</i>			
<i>AD Law</i>	-0.007** (-2.099)	-0.017*** (-3.807)	-0.008** (-2.341)	-0.008** (-2.367)
<i>AD Law Neighboring</i>				0.003 (0.658)
Control variables	Yes	Yes	No	Yes
Firm and year fixed effects	Yes	Yes	Yes	Yes
SIC2-year fixed effects	Yes	No	No	No
Census division-year fixed effects	No	Yes	No	No
Beg characteristic-year fixed effects	No	No	Yes	No
R-squared	0.74	0.71	0.94	0.71
Observations	11,933	13,223	13,154	13,235

Table 6: Effects of AD laws on leverage

Ordinary least squares regressions for a sample of public firms from 1946 to 1964. *AD Law* is a dummy set to one if a state-level anti-discrimination law is in place, and zero otherwise. The dependent variable in Panel A is *Book Leverage* which is the sum of debt in current liabilities and long-term debt (Compustat DLC plus DLTT) divided by the book value of assets (Compustat AT).. In Panel B, the dependent variable is *Total Leverage* which is total liabilities (Compustat LT) divided by the book value of assets (Compustat AT). In Panel C, the dependent variable is *Market Leverage* which is the sum of debt in current liabilities and long-term debt (Compustat DLC plus DLTT) divided by the market value of assets (Compustat items DLC+DLTT + CSHO*PRCC_F). All other variables are described in the Appendix. All regressions include firm fixed effects and year fixed effects. Column 3 also includes state-specific time trends. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

	(1)	(2)	(3)
<i>Panel A: Book Leverage</i>			
Dep. variable	<i>Book Leverage</i>	<i>Book Leverage</i>	<i>Book Leverage</i>
<i>AD Law</i>	-0.013** (-2.539)	-0.015*** (-3.146)	-0.014*** (-2.86)
<i>Log (Total Assets)</i>		0.075*** (10.193)	0.077*** (10.67)
<i>Return on Assets</i>		-0.709*** (-16.164)	-0.703*** (-16.30)
<i>Fixed Assets</i>		0.117*** (3.765)	0.124*** (4.08)
<i>Dividend Payer</i>		-0.027*** (-4.282)	-0.027*** (-4.29)
<i>State Income Growth</i>		0.000 (0.666)	0.000 (0.90)
Firm and year fixed effects	Yes	Yes	Yes
State-specific time trends	No	No	Yes
Observations	12,798	12,721	12,721
R-squared	0.79	0.83	0.83
<i>Panel B: Total Leverage</i>			
Dependent variable	<i>Total Leverage</i>	<i>Total Leverage</i>	<i>Total Leverage</i>
<i>AD Law</i>	-0.019*** (-3.204)	-0.021*** (-3.618)	-0.016*** (-3.01)
<i>Log (Total Assets)</i>		0.073*** (8.967)	0.072*** (8.88)
<i>Return on Assets</i>		-0.387*** (-7.916)	-0.379*** (-7.87)
<i>Fixed Assets</i>		-0.052 (-1.598)	-0.047 (-1.45)
<i>Dividend Payer</i>		-0.021*** (-3.313)	-0.022*** (-3.40)
<i>State Income Growth</i>		0.001*** (2.828)	0.001*** (2.69)
Firm and year fixed effects	Yes	Yes	Yes
State-specific time trends	No	No	Yes
Observations	12,490	12,406	12,406
R-squared	0.81	0.83	0.83

continued...

Table 6, continued

	(1)	(2)	(3)
<i>Panel C: Market Leverage</i>			
Dep. variable	<i>Market Leverage</i>	<i>Market Leverage</i>	<i>Market Leverage</i>
<i>AD Law</i>	-0.020** (-2.115)	-0.020** (-2.517)	-0.020*** (-2.59)
<i>Log (Total Assets)</i>		0.081*** (6.762)	0.083*** (6.83)
<i>Return on Assets</i>		-1.339*** (-15.424)	-1.325*** (-15.34)
<i>Fixed Assets</i>		0.162*** (3.347)	0.172*** (3.59)
<i>Dividend Payer</i>		-0.050*** (-3.950)	-0.047*** (-3.73)
<i>State Income Growth</i>		0.000 (0.321)	0.001 (0.93)
Firm and year fixed effects	Yes	Yes	Yes
State-specific time trends	No	No	Yes
Observations	7,643	7,625	7,625
R-squared	0.77	0.82	0.83

Table 7: Robustness tests on the effects of AD laws on leverage

Ordinary least squares regressions for a sample of public firms from 1946 to 1964. *AD Law* is a dummy set to one if a state-level anti-discrimination law is in place, and zero otherwise. The dependent variable is *Book Leverage* which is the sum of debt in current liabilities and long-term debt (Compustat *DLC* plus *DLTT*) divided by the book value of assets (Compustat *AT*). Column 1 includes two digit SIC-year fixed effects, column 2 includes Census division-year fixed effects, and column 3 includes beginning characteristic-year fixed effects. Column 4 includes *AD Law Neighboring*, which is the median value of *AD Law* status based on neighboring states. All regressions include firm fixed effects and year fixed effects as well as controls as in Table 6. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

Dep. Variable	(1)	(2)	(3)	(4)
	<i>Book Leverage</i>			
<i>AD Law</i>	-0.014*** (-2.774)	-0.011* (-1.911)	-0.014*** (-2.605)	-0.014*** (-2.92)
<i>AD Law Neighboring</i>				-0.003 (-0.48)
Control variables	Yes	Yes	No	Yes
Firm and year fixed effects	Yes	Yes	Yes	Yes
SIC2-year fixed effects	Yes	No	No	No
Census division-year fixed effects	No	Yes	No	No
Beg characteristic-year fixed effects	No	No	Yes	No
R-squared	0.85	0.83	0.91	0.83
Observations	11,307	12,711	12,469	12,721

Table 8: Robustness to dropping regions/states and strength of AD laws

Ordinary least squares regressions for a sample of public firms from 1946 to 1964. *Operating Profitability* is the ratio of Compustat *OIBDP* to *AT*. *Book Leverage* is the sum of debt in current liabilities and long-term debt (Compustat *DLC* plus *DLTT*) divided by the book value of assets (Compustat *AT*). *AD Law* is a dummy set to one if a state-level anti-discrimination law is in place, and zero otherwise. Regressions include control variables, firm fixed effects, year fixed effects, and state-specific time trends. *RTW* is a dummy variable equal to 1 if a state has a right-to-work law in place, and zero otherwise. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

Dep. Variable:	(1) <i>Operating Profitability</i>	(2) <i>Book Leverage</i>
<i>Panel A: Control for RTW Laws</i>		
<i>AD Law</i>	-0.011*** (-3.230)	-0.014*** (-2.893)
<i>RTW</i>	-0.025 (-1.123)	-0.066** (-2.301)
Control variables	Yes	Yes
Firm and year fixed effects	Yes	Yes
State-specific time trends	Yes	Yes
Observations	13,235	12,721
<i>Panel B: Exclude South</i>		
<i>AD Law</i>	-0.011*** (-3.298)	-0.014*** (-2.866)
Control variables	Yes	Yes
Firm and year fixed effects	Yes	Yes
State-specific time trends	Yes	Yes
Observations	11,166	10,745
<i>Panel C: Exclude New York</i>		
<i>AD Law</i>	-0.007** (-2.105)	-0.007* (-1.667)
Control variables	Yes	Yes
Firm and year fixed effects	Yes	Yes
State-specific time trends	Yes	Yes
Observations	10,938	10,479
<i>Panel D: Exclude Illinois</i>		
<i>AD Law</i>	-0.015*** (-3.632)	-0.018*** (-3.119)
Control variables	Yes	Yes
Firm and year fixed effects	Yes	Yes
State-specific time trends	Yes	Yes
Observations	11,757	11,300

Table 9: Augmented Sample Results

Ordinary least squares regressions for an augmented sample of public firms from 1946 to 1964. The augmented sample is based on Landes (1968). *AD Law Augmented* is a dummy set to one if a state-level anti-discrimination law is in place, and zero otherwise. *AD Law Enforce* is a dummy set to one if a state-level anti-discrimination law is in place and the state commission had enforcement powers, and zero otherwise. *AD Law Non Enforce* is a dummy set to one if a state-level anti-discrimination law is in place and the state commission did not have enforcement powers, and zero otherwise. *Operating Profitability* is the ratio of Compustat *OIBDP* to *AT*. *Book Leverage* is the sum of debt in current liabilities and long-term debt (Compustat *DLC* plus *DLTT*) divided by the book value of assets (Compustat *AT*). Regressions include control variables, firm fixed effects, year fixed effects, and state-specific time trends. *RTW* is a dummy variable equal to 1 if a state has a right-to-work law in place, and zero otherwise. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

Dep. Variable:	(1) <i>Operating Profitability</i>	(2) <i>Book Leverage</i>
<i>Panel A: Augmented Sample</i>		
<i>AD Law Augmented</i>	-0.010*** (-3.114)	-0.011** (-2.444)
Control variables	Yes	Yes
Firm and year fixed effects	Yes	Yes
State-specific time trends	Yes	Yes
Observations	13,629	13,108
<i>Panel B: Strength of AD Law</i>		
<i>AD Law Enforce</i>	-0.011*** (-3.179)	-0.012** (-2.566)
<i>AD Law Non Enforce</i>	0.004 (0.258)	0.020 (1.413)
Control variables	Yes	Yes
Firm and year fixed effects	Yes	Yes
State-specific time trends	Yes	Yes
Observations	13,629	13,108

Table 10: Heterogeneity analyses based on state-level percent Black population

Ordinary least squares regressions for a sample of public firms from 1946 to 1964. High/low sub-samples are created based on above/below median state-level percent Black population in 1950. *AD Law* is a dummy set to one if a state-level anti-discrimination law is in place, and zero otherwise. *Operating Profitability* is the ratio of Compustat *OIBDP* to *AT*. *Book Leverage* is the sum of debt in current liabilities and long-term debt (Compustat *DLC* plus *DLTT*) divided by the book value of assets (Compustat *AT*). All estimations contain firm fixed effects, year fixed effects, and state-specific time trends. All other variables are described in the Appendix. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

Dep. Variable:	(1)	(2)	(3)	(4)
	<i>Operating Profitability</i>		<i>Book Leverage</i>	
	High	Low	High	Low
<i>AD Law</i>	-0.018*** (-3.166)	-0.003 (-0.587)	-0.023*** (-3.270)	-0.008 (-1.317)
<i>Log (Total Assets)</i>	-0.000 (-0.045)	0.013** (2.235)	0.076*** (8.249)	0.078*** (6.825)
<i>Fixed Assets</i>	-0.144*** (-5.469)	-0.124*** (-3.861)	0.169*** (4.408)	0.061 (1.244)
<i>Dividend Payer</i>	0.029*** (4.752)	0.020*** (3.168)	-0.034*** (-4.448)	-0.013 (-1.277)
<i>State Income Growth</i>	0.002*** (4.352)	0.002*** (3.191)	0.002*** (3.276)	-0.001** (-2.357)
<i>Return on Assets</i>			-0.708*** (-13.613)	-0.705*** (-8.984)
Difference between coefficients	-0.015**		-0.015*	
Firm and year fixed effects	Yes	Yes	Yes	Yes
State-specific time trends	Yes	Yes	Yes	Yes
Observations	7,965	5,268	7,747	4,972
R-squared	0.70	0.73	0.84	0.82

Table 11: Heterogeneity analyses based on city-level percent nonwhite population

Ordinary least squares regressions for a sample of public firms from 1946 to 1964. High/low sub-samples are created based on above/below median city-level (SMA-level) percent nonwhite population in 1950. *AD Law* is a dummy set to one if a state-level anti-discrimination law is in place, and zero otherwise. *Operating Profitability* is the ratio of Compustat *OIBDP* to *AT*. *Book Leverage* is the sum of debt in current liabilities and long-term debt (Compustat *DLC* plus *DLTT*) divided by the book value of assets (Compustat *AT*). All estimations contain firm fixed effects, year fixed effects, and state-specific time trends. All other variables are described in the Appendix. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

Dep. Variable:	(1)	(2)	(3)	(4)
	<i>Operating Profitability</i>		<i>Book Leverage</i>	
	High	Low	High	Low
<i>AD Law</i>	-0.016*** (-2.924)	-0.002 (-0.481)	-0.025*** (-3.522)	-0.007 (-1.197)
<i>Log (Total Assets)</i>	-0.002 (-0.420)	0.014** (2.085)	0.076*** (8.192)	0.078*** (6.910)
<i>Fixed Assets</i>	-0.136*** (-5.320)	-0.111*** (-3.083)	0.142*** (3.759)	0.066 (1.233)
<i>Dividend Payer</i>	0.025*** (4.008)	0.028*** (4.325)	-0.036*** (-4.582)	-0.012 (-1.008)
<i>State Income Growth</i>	0.002*** (3.969)	0.002*** (4.003)	0.002*** (2.924)	-0.001** (-2.334)
<i>Return on Assets</i>			-0.721*** (-13.831)	-0.665*** (-7.744)
Difference between coefficients	-0.014**		-0.018*	
Firm and year fixed effects	Yes	Yes	Yes	Yes
State-specific time trends	Yes	Yes	Yes	Yes
Observations	7,428	4,919	7,235	4,643
R-squared	0.70	0.73	0.83	0.83

Table 12: Additional heterogeneity analyses

Ordinary least squares regressions for a sample of public firms from 1946 to 1964. *AD Law* is a dummy set to one if a state-level anti-discrimination law is in place, and zero otherwise. *Operating Profitability* is the ratio of Compustat *OIBDP* to *AT*. *Book Leverage* is the sum of debt in current liabilities and long-term debt (Compustat *DLC* plus *DLTT*) divided by the book value of assets (Compustat *AT*). Regressions include control variables, firm fixed effects, year fixed effects, and state-specific time trends. In Panel A, sub-samples are based on labor intensity which is measured as *Emp/Sales* (where *Sales* is revenues and *Emp* is number of employees as obtained from Compustat or Moodys manuals). Panel B is based on sub-samples of African American migration which is measured as the growth rate in the number of Southern-born African Americans living in each state between 1950 and 1970. In Panel C, industry concentration is defined using the Herfindahl-Hirschman Index (HHI) calculated using revenue-based market share in the two-digit SIC industry of firms. The high and low split is based on sample median in all three panels. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

	(1)	(2)	(3)	(4)
Dep. Variable:	<i>Operating Profitability</i>		<i>Book Leverage</i>	
<i>Panel A: Split by labor intensity</i>	High	Low	High	Low
<i>AD Law</i>	-0.011*** (-2.236)	-0.005 (-0.936)	-0.025*** (-3.465)	-0.003 (-0.391)
Control variables	Yes	Yes	Yes	Yes
Firm and year fixed effects	Yes	Yes	Yes	Yes
State-specific time trends	Yes	Yes	Yes	Yes
Observations	5,291	5,296	5,065	5,058
Difference between coefficients		-0.006		-0.022**
<i>Panel B: Split by African American migration</i>	High	Low	High	Low
<i>AD Law</i>	-0.022*** (-3.911)	0.004 (0.865)	-0.017** (-2.289)	-0.003 (-0.445)
Control variables	Yes	Yes	Yes	Yes
Firm and year fixed effects	Yes	Yes	Yes	Yes
State-specific time trends	Yes	Yes	Yes	Yes
Observations	7,099	4,049	6,794	3,932
Difference between coefficients		-0.018***		-0.014*
<i>Panel C: Split by industry concentration</i>	High	Low	High	Low
<i>AD Law</i>	-0.015*** (-3.381)	-0.009* (-1.908)	-0.023*** (-3.188)	0.000 (0.006)
Control variables	Yes	Yes	Yes	Yes
Firm and year fixed effects	Yes	Yes	Yes	Yes
State-specific time trends	Yes	Yes	Yes	Yes
Observations	6,326	6,377	6,161	6,019
Difference between coefficients		-0.006		-0.023**

How do anti-discrimination laws affect firm performance and financial policies? Evidence from the post-World War II period

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Internet Appendix A: Determinants of the timing of anti-discrimination laws

To investigate how state-level macroeconomic trends affect the timing of AD laws, we estimate Cox proportional hazard models. We adopt state-level demographic and economic characteristics studied by Collins (2003b) and employ hazard models where we fix covariates at a given point or time, as well as models where we employ time-varying covariates (subject to data availability). We examine the following demographic variables following Collins (2003b). First, we study the percentage of the state's population that is Black. In the fixed covariate models, this variable is measured based on the 1950 Census data. For the time-varying covariate model, we linearly interpolate the percentage of Black population using the 1940, 1950, and 1960 Census data. Second, to assess the impact of religious organizations in the state, we measure the Jewish and Catholic proportion of each state's population in 1952. We are unable to obtain this data in a time-varying fashion. Third, we calculate separate measures to capture the percentage of each state's population that is affiliated with the AFL and CIO unions in 1953. This data is collected from Troy (1957) and is available only at a fixed point in time.

To measure state economic conditions, we employ two main variables. First, we include the overall unemployment rate in the state. Alternatively, following Collins (2003b), instead of measuring unemployment across all races, we measure the unemployment rate among whites. In the fixed covariate model, unemployment is measured based on the 1950 Census data and for the time-varying covariate model, it is interpolated using the 1940, 1950, and 1960 Census data. Collins (2003b) argues that linear interpolation of unemployment rates during war years is infeasible. He forces each state's unemployment to reach the national average for years 1943, 1944, and 1945, and then continues with linear interpolation between 1945 and 1950 and then between 1950 and 1960. We implement a similar interpolation strategy as Collins (2003b). Second, we include the growth rate in the state-level real per capita personal income. Annual data is available on this variable from the Bureau of Economic Analysis. For the fixed covariate model, we measure this variable in year 1950. We use annual data on personal income for the time-varying covariate model. Overall, if deteriorating state-level economic conditions are driving AD law adoption, we should see that AD laws were implemented sooner in states with higher unemployment rates and lower growth in real per capita income. Alternatively, if greater economic growth led states to adopt AD laws sooner then we should see opposite results.

The results from the hazard model are presented in Internet Appendix Table 1. The sample comprises only those states that appear in our firm-year panel for which we are able to collect data on the above variables. The dependent variable is the number of years from 1940 to the year of adoption of the AD law. Hazard ratios are reported in the table. Hazard ratios greater than 1 indicate that an increase in that variable leads to faster adoption of the law. In Columns 1 and 2, we report results from hazard models based on fixed-covariates, and in Columns 3 and 4, we report results based on time-varying covariate hazard models. Our time-varying covariates, namely the percentage of each state's population that is Black, unemployment rate (overall and White), and the growth rate in the state-level real per capita personal income, are denoted with superscript mark ^(*). In all four columns, we find that the coefficients on overall and White unemployment rate are statistically insignificant. Similarly, we find that the coefficients on the growth in real per capita personal income are also statistically insignificant. These findings allay concerns that state-level macroeconomic conditions are the drivers of AD law adoption. Finally, the signs on demographic variables are generally consistent with those found in Collins (2003b). Specifically, states with higher Jewish and Catholic affiliation are likely to adopt AD laws faster. Similarly, states with higher affiliation to AFL and CIO unions adopt AD laws faster, and states with a higher percentage of Black population are slower to adopt AD laws.

Internet Appendix Table 1: Determinants of timing of anti-discrimination laws (Cox proportional hazard model)

The table reports hazard ratios from Cox proportional hazard models estimating the timing of AD law adoption. z-Statistics are in parentheses and indicate whether the hazard ratios are different from 1. Columns 1 and 2 contain results from fixed covariate models, while Columns 3 and 4 contains results from time-varying covariate models. The time varying covariates are denoted by superscript “^a” and include four variables: *Unemployment rate*, *State Income Growth*, *White Unemployment rate*, and *% Black*. *Unemployment rate* is the state-level unemployment rate based on Census data. *Growth in Real Per Capita Income* is annual growth rate in the state-level real per capita income. *White Unemployment rate* is the state-level unemployment rate for Whites based on Census data. *% Jewish* (*% Catholic*) is the percentage of the state’s population that is Jewish (Catholic). *% Black* is the percentage of the state’s population that was Black. *% CIO Union* (*% AFL Union*) is the percentage of the state’s population affiliated with the CIO (AFL) union in 1953. Columns 1 and 2 use robust standard errors and Columns 3 and 4 use robust standard errors clustered by state. Further details on variable construction are provided in Internet Appendix A. The symbols ***, **, and * * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

	(1)	(2)	(3)	(4)
	Fixed covariates		Time-varying covariates	
<i>Unemployment rate</i> ^a	0.970 (-0.242)		0.995 (-0.424)	
<i>White Unemployment rate</i> ^a		0.977 (-0.181)		0.995 (-0.346)
<i>State Income Growth</i> ^a	1.067 (0.512)	1.068 (0.520)	0.998 (-0.400)	0.998 (-0.411)
<i>% Jewish</i>	1.320*** (3.043)	1.319*** (3.047)	1.286*** (3.345)	1.285*** (3.345)
<i>% Catholic</i>	1.049** (2.288)	1.049** (2.280)	1.053*** (2.779)	1.053*** (2.778)
<i>% Black</i> ^a	0.823* (-1.874)	0.821* (-1.950)	0.857** (-1.968)	0.857** (-1.972)
<i>% CIO Union</i>	1.103* (1.645)	1.104* (1.649)	1.106** (1.978)	1.106** (1.978)
<i>% AFL Union</i>	1.102** (2.113)	1.102** (2.104)	1.094** (2.268)	1.094** (2.267)
Number of states	34	34	34	34

Internet Appendix Table 2: Anti-discrimination laws and leverage, robustness to alternative measures

Ordinary least squares regressions for a sample of public firms from 1946 to 1964. *AD Law* is a dummy set to one if a state-level anti-discrimination law is in place, and zero otherwise. *Book Leverage* is the sum of debt in current liabilities and long-term debt (Compustat items *DLC* plus *DLTT*) divided by the book value of assets (*AT*). *Total Leverage* is total liabilities (*LT*) divided by the book value of assets (*AT*). *Market Leverage* is the sum of debt in current liabilities and long-term debt (*DLC* plus *DLTT*) divided by the market value of assets (*DLC+DLTT + CSHO*PRCC_F*). *Operating Profitability* is *OIBDP* divided by *AT*. *Log(1+Total Debt)* is the natural log of one plus debt in current liabilities and long-term debt ($\ln(1+DLC+DLTT)$). Net measures simply subtract cash from the numerator. *Net Book Leverage* is $(DLC+DLTT-CHE)/AT$. *Net Market Leverage* is $(DLC+DLTT-CHE)/(DLC+DLTT + CSHO*PRCC_F)$. *Earnings Volatility* is the standard deviation of *NI/AT* over the past 10 years (minimum of three years of data required). All other variables are described in the Appendix. All regressions include firm fixed effects, year fixed effects, and state-specific trends. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * * denote $p<0.01$, $p<0.05$, and $p<0.1$ respectively.

	(1)	(2)	(3)	(4)	(4)	(5)	(6)
Dep. Variable:	<i>Book Leverage</i>	<i>Total Leverage</i>	<i>Market Leverage</i>	<i>Operating Profitability</i>	<i>Log(1+Total Debt)</i>	<i>Net Book Leverage</i>	<i>Net Market Leverage</i>
<i>AD Law</i>	-0.014*** (-2.819)	-0.019*** (-3.328)	-0.021*** (-2.716)	-0.011*** (-3.124)	-0.073* (-1.717)	-0.012* (-1.943)	-0.019* (-1.689)
<i>Log (Total Assets)</i>	0.089*** (10.064)	0.089*** (9.536)	0.090*** (7.094)	0.011** (2.240)	1.102*** (20.709)	0.094*** (9.180)	0.111*** (5.836)
<i>Return on Assets</i>	-0.746*** (-13.294)	-0.468*** (-7.721)	-1.390*** (-13.750)		-4.991*** (-12.771)	-1.060*** (-15.020)	-1.233*** (-10.567)
<i>Fixed Assets</i>	0.117*** (3.146)	-0.074* (-1.931)	0.154*** (3.260)	-0.129*** (-5.709)	0.751*** (3.614)	0.457*** (10.265)	0.515*** (7.819)
<i>Dividend Payer</i>	-0.024*** (-3.058)	-0.013* (-1.661)	-0.054*** (-3.781)	0.020*** (3.681)	-0.076* (-1.649)	-0.027*** (-2.762)	-0.045** (-2.534)
<i>State Income Growth</i>	0.000 (1.352)	0.001** (2.012)	0.001* (1.808)	0.002*** (4.394)	0.004 (1.172)	0.001 (1.490)	0.001 (1.157)
<i>Earnings Volatility</i>	0.656*** (3.430)	0.710*** (3.516)	0.302 (1.268)	-0.646*** (-4.913)	1.806 (1.511)	0.529** (2.317)	0.635* (1.843)
Firm and year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-specific time trends	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9,425	9,185	6,466	9,810	9,425	9,413	6,454
R-squared	0.84	0.84	0.84	0.74	0.92	0.87	0.81

Internet Appendix Table 3: Propensity score matching

Ordinary least squares regressions using a propensity score matched sample of public firms over a period of five years around the passage of *AD Law*. Treatment firms are those in states where an *AD Law* was adopted in year t whereas control firms are those where no *AD law* is in place. We require that each treatment and control firm have at least one year of data available in the five years prior to and five years after the adoption. We run a logistic regression model where we predict the likelihood of being a treated firm. The explanatory variables in the logistic regression are total assets, net income margin, and fixed assets when finding matches for *Book Leverage* regression. We do not include net income margin when finding matches for the *Operating Profitability* regression since the two would be highly correlated. We then match each treatment observation to a control observation with replacement in the same year and the closest propensity score calculated from the logistic regression. The sample comprises all observations of the treatment and matched control firms in the five years before and five years after the adoption of the laws. *Treatment* is a dummy set to one for firms with a state-level anti-discrimination law (*AD Law*) is in place, and zero for matched control firms. *Post* is a dummy set to one for years after the passage of *AD Law*, and set to zero otherwise. All variables are described in the Appendix. All models contain firm fixed effects and year fixed effects as well as state-specific trends. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

Panel A: Differences between treatment and matched control firms

	Treatment	Control	Diff.
<i>Log (Total Assets)</i>	4.068	4.084	-0.016 (-0.16)
<i>Fixed Assets</i>	0.379	0.364	0.015 (1.13)
<i>Net Income Margin</i>	0.067	0.064	0.003 (1.24)
<i>Dividend Payer</i>	0.881	0.881	0.00 (1.00)
<i>Book Leverage</i>	0.165	0.176	-0.011 (-1.12)

Panel B: Multivariate Analysis

Dep. Variable:	(1) <i>Book Leverage</i>	(2) <i>Book Leverage</i>	(3) <i>Operating Profitability</i>	(4) <i>Operating Profitability</i>
<i>Treatment x Post</i>	-0.008* (-1.677)	-0.011*** (-2.696)	-0.005 (-1.613)	-0.006* (-1.686)
<i>Post</i>	0.005 (1.330)	0.003 (0.896)	0.001 (0.437)	0.001 (0.296)
Control variables	No	Yes	No	Yes
Firm and year fixed effects	Yes	Yes	Yes	Yes
State-specific time trends	Yes	Yes	Yes	Yes
Observations	7,103	7,094	7,471	7,466
R-Squared	0.78	0.83	0.73	0.74

Internet Appendix Table 4: Timing of law passage

Ordinary least squares regressions for a sample of public firms from 1946 to 1964. *Book Leverage* is the sum of debt in current liabilities and long-term debt (Compustat *DLC* plus *DLTT*) divided by the book value of assets (Compustat *AT*). *Operating Profitability* is operating profitability measured as the ratio of Compustat *OIBDP* over *AT*. *AD Law*⁻² is a dummy variable set to one for years where the law was passed in two years and zero otherwise, *AD Law*⁻¹ is a dummy variable set to one for years where the law was passed the next year and zero otherwise, *AD Law*⁰ is a dummy variable set to one for years where the law was passed in the current year and zero otherwise, *AD Law*¹ is a dummy variable set to one to indicate years where the law was passed the previous year and zero otherwise, and *AD Law*²⁺ is a dummy variable set to one to indicate years where the law was passed two or more years ago and zero otherwise. All other variables are described in the Appendix. All models contain firm fixed effects and year fixed effects as well as state-specific trends. Standard errors reported in the parentheses are based on heteroscedasticity robust standard errors that are clustered at the firm level. The symbols ***, **, and * * denote $p < 0.01$, $p < 0.05$, and $p < 0.1$ respectively.

Dep. Variable:	(1) <i>Book Leverage</i>	(2) <i>Operating Profitability</i>
<i>AD Law</i> ⁻²	-0.005 (-1.051)	-0.002 (-0.726)
<i>AD Law</i> ⁻¹	-0.010** (-2.160)	-0.004 (-0.900)
<i>AD Law</i> ⁰	-0.014** (-2.435)	-0.008* (-1.716)
<i>AD Law</i> ¹	-0.023*** (-3.218)	-0.008 (-1.599)
<i>AD Law</i> ²⁺	-0.019*** (-2.706)	-0.015*** (-2.832)
<i>Log (Total Assets)</i>	0.077*** (10.678)	0.005 (1.249)
<i>Fixed Assets</i>	0.124*** (4.083)	-0.138*** (-6.730)
<i>Dividend Payer</i>	-0.027*** (-4.280)	0.026*** (5.697)
<i>State Income Growth</i>	0.000 (0.354)	0.002*** (5.005)
<i>Return on Assets</i>	-0.703*** (-16.296)	
Firm and year fixed effects	Yes	Yes
State-specific time trends	Yes	Yes
Observations	12,721	13,235
R-Squared	0.83	0.71

Internet Appendix Table 5: Falsification (randomization) tests

This table shows summary statistics from a falsification exercise where we assign each state a different year of adoption while maintaining the distribution of adoption dates in Table 1. Thus, two states are randomly assigned an adoption year of 1945, one state is randomly assigned an adoption year of 1946, etc., and thirteen states are randomly assigned no adoption year. Next, we replicate the firm fixed effects panel regressions with *Operating Profitability* and *Book Leverage* as our dependent variables and an AD law dummy based on randomly assigned adoption dates as our key independent variable. Specifically, we use the specifications in Table 4, Panel A, Column 3 and Table 6, Panel A, Column 3. We repeat this process a total of 100 times and the table below shows the distribution of the coefficient on the AD law dummy, the standard error, and the t-statistic.

Dep. Variable:	<i>Operating Profitability</i>	<i>Book Leverage</i>
<u>Randomized AD law dummy</u>		
Coefficient, mean	-0.001	-0.002
Coefficient, median	0.0002	-0.002
Standard error, mean	0.005	0.006
Standard error, median	0.004	0.006
t-Statistic, mean	-0.158	-0.252
t-Statistic, median	0.042	-0.287
Number of replications	100	100
Control Variables	Yes	Yes
Firm and year fixed effects	Yes	Yes
State-specific time trends	Yes	Yes
Number of observations per replication	13,235	12,721